

SIVA NAGA RAJU PARIMISSETTY

Full Stack AI/ML Engineer | Python & Java Developer | Cloud & DevOps Specialist

+1 (267) 309-1872 | sivaparimisetty849@gmail.com | USA | linkedin.com/in/siva-naga-raju-parimisetty

OBJECTIVE

- Results-driven Full Stack AI/ML Engineer with 8+ years of experience architecting and delivering production-grade AI systems, Python/Java backends, React/Node.js frontends, and scalable cloud-native services on AWS spanning healthcare, fintech, and graphics engineering domains.
- Expert in end-to-end ML engineering lifecycle: model design, training, hyperparameter tuning, deployment on AWS SageMaker, real-time inference optimization, and MLOps automation via CI/CD pipelines (AWS CodePipeline, GitHub Actions, GitLab CI).
- Proven track record building and securing GenAI pipelines, RAG systems, encrypted embeddings, prompt-injection guardrails, and LLM inference APIs using LangChain, LlamaIndex, and OpenAI API for enterprise product teams.
- Deep expertise in microservices architecture using Docker, Kubernetes, and AWS ECS Fargate, achieving up to 60% latency reduction through async FastAPI design, connection pooling, and tail-latency optimization.
- Proficient across a broad technology stack: C++, Java, Python, JavaScript (ES2022), TypeScript, HTML5, Node.js, React, Next.js, SQL, and Bash delivering robust, cross-functional solutions from API layer to UI.
- Committed to engineering excellence: PHI-compliant ML pipelines, HIPAA-aligned data handling, zero-downtime canary deployments, and measurable business outcomes at scale.

WORK EXPERIENCE

Full Stack AI Engineer | Morgan Stanley | USA | May 2025 – Present

- Architected and deployed GenAI summarization and retrieval microservices on AWS ECS Fargate using FastAPI (Python), integrating securely into risk and research dashboards with end-to-end encrypted data flows — reducing analyst query time by 40%.
- Designed and exposed a strongly typed Flask inference API over a unified AWS S3 data lake, consolidating historical trade and research data consumed by multiple internal analytics tools — eliminating data silos and improving data accessibility.
- Led decomposition of a monolithic Python backend into domain-focused FastAPI microservices, standardizing authentication (JWT), request validation (Pydantic), and structured observability (CloudWatch) across all GenAI workloads.
- Extended a RAG pipeline for a portfolio-assistant chatbot with multi-layer guardrails against prompt injection and data leakage, enforcing policy at both API gateway and model inference layers using LangChain and custom middleware.
- Redesigned I/O architecture with async FastAPI endpoints and PostgreSQL connection pooling, reducing tail latency by 180 ms (37% improvement) under high-throughput model inference scenarios.
- Implemented retrieval over encrypted embeddings in AWS (Pinecone + KMS) to satisfy strict data residency and compliance mandates, ensuring GenAI responses never expose client-identifying attributes.
- Automated end-to-end CI/CD for AI microservices via AWS CodePipeline with containerized test stages, enabling zero-downtime canary rollouts across 4 production clusters — cutting release cycle time by 50%.

ML Engineer | UnitedHealth Group | Remote | Oct 2023 – May 2025

- Centralized model inference by migrating 12+ scoring pipelines from legacy services to Python microservices on AWS SageMaker with API Gateway contract versioning, increasing reliability and eliminating deployment conflicts across teams.
- Designed and owned the core FastAPI inference API surface for clinical risk models (Java + Python hybrid architecture), maintaining backward-compatible Pydantic request schemas across multiple internal model iterations serving 12 million+ monthly predictions.
- Engineered an S3-driven batch scoring path integrated into existing claims workflows during a release freeze, achieving zero downstream format changes while unblocking a critical business deliverable.
- Optimized SageMaker endpoint configurations and model input pipelines, slashing median end-to-end prediction latency from 480 ms to 210 ms — a 56% performance improvement directly impacting clinician decision support tools.
- Developed a PHI-compliant masked training data pipeline using AWS Lambda and S3 (Python), feeding nightly retraining jobs in SageMaker while maintaining full HIPAA compliance across all data handling touchpoints.
- Established real-time model monitoring using CloudWatch dashboards and alerts, tuning class thresholds and feature scaling in Python/scikit-learn to stabilize precision and recall across production clinical risk models.

Application Engineer (Full Stack) | NVIDIA Graphics Pvt. Ltd. | India | Sep 2021 – Jan 2023

- Designed and delivered a unified React + Node.js interface backed by FastAPI (Python) microservices to consolidate fragmented graphics configuration workflows — reducing engineer time-on-task by 35% and eliminating manual configuration errors.
- Built robust Python FastAPI backend services for shader configuration, driver status monitoring, and GPU telemetry, incorporating Pydantic request validation, JWT auth middleware, structured logging, and Swagger/OpenAPI documentation.

- Refactored a tightly coupled React UI that queried GPUs directly into clean REST APIs with strong contracts, enabling consistent validation, predictable latency, and testability for real-time graphics workloads.
- Applied in-process caching (Redis) and batched GPU metadata queries to API handlers, cutting median response latency from 240 ms to 95 ms — a 60% reduction that sustained frame-time stability under peak graphics load.
- Instrumented React views and backend endpoints with performance metrics pre-release using Jest and React Testing Library, identifying and tuning hot paths to sustain stable throughput under load-test traffic.

Software Engineer | Kloc Technologies (Nvidia) | India | Apr 2018 – Sep 2021

- Developed and maintained RESTful API endpoints in PHP, Java, and JavaScript for enterprise backend systems, aligning validation, error-handling, and logging logic with industry coding standards and team conventions.
- Refactored an untested JavaScript dashboard module into modular, testable components; introduced comprehensive unit test coverage using Jest, eliminating recurring UI regressions and improving deployment confidence.
- Diagnosed and resolved intermittent API failures traced to faulty MySQL joins and N+1 query patterns, reducing production error incidents by approximately 40 per day and improving application stability.
- Delivered full-stack feature development across HTML5/CSS3 frontend components and backend handler integration using Node.js and Java, resolving multiple front-to-back integration defects and improving end-user experience.
- Collaborated cross-functionally with QA and product teams using Git-based workflows, code reviews, and Postman API testing to maintain release quality across a high-velocity delivery environment.

EDUCATION

Master of Science in Computer Science

Lewis University, Romeoville, IL
Jan 2023 May 2024

Bachelor of Science in Computer Science

Gudlavalleru Engineering College, Andhra Pradesh, India
Mar 2014 – Jun 2018

CERTIFICATIONS

AWS Certified Developer – Associate

Amazon Web Services
Jan 2026

TECHNICAL SKILLS

Backend Development: Python, Django, Flask, FastAPI, REST APIs, Microservices, Event-Driven Architecture

Frontend Development: React, Angular, Vue.js, Full Stack Development

Cloud & DevOps: AWS, Azure, GCP, Red Hat OpenStack, OpenStack Operators, Helm Charts, Docker, Kubernetes, OpenShift, CI/CD Pipelines, GitHub Actions, GitLab, Jenkins, Azure DevOps

GenAI & LLM Technologies: Generative AI, RAG, Embeddings, Vector Search, Retrieval Tuning, Prompt Engineering, OpenAI, Azure OpenAI, Anthropic, Hugging Face, LangChain, LlamaIndex

Vector Databases & Document Processing: FAISS, Milvus, Pinecone, pgvector, PDF Processing, HTML Processing, Mark-down Processing, Chunking Strategies

Databases & Caching: PostgreSQL, MongoDB, Redis

Security & Integration: OAuth2, OIDC, JWT, API Security, IAM, Policy-Based Authorization, PlainID

Messaging & Networking: Kafka, SQS, PubSub, Networking

Testing & Monitoring: Unit Testing, Integration Testing, E2E Testing, Observability, Performance Optimization

Collaboration & Documentation: Communication, Design Documentation, Stakeholder Collaboration

KEY ACHIEVEMENTS & IMPACT

- 56% reduction in ML model inference latency at UnitedHealth Group (480 ms → 210 ms) — directly improving clinician decision support at scale.
- 60% API response latency reduction at NVIDIA (240 ms → 95 ms) — enabling real-time GPU telemetry under frame-time constraints.
- 180 ms tail latency reduction at Morgan Stanley via async FastAPI redesign — improving GenAI service SLAs under high-load inference.
- 50% faster release cycles through AWS CodePipeline CI/CD automation with canary deployments across 4 production clusters.
- 40% reduction in analyst query time by consolidating fragmented data systems into a unified GenAI retrieval architecture.
- HIPAA-compliant ML pipeline serving 12 million+ monthly clinical risk predictions with measurable precision/recall stability.